Design Pattern Final Project

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Ontology and Intel Analysis

Revised October 17, 2026: proposal re-examined against course material

Presentation 2, November 11, 2026: implemented, tested, extendable

Proposed Final Project Topic

Meeting

Why Meetings?

- Meeting presentations and meeting minutes are our main target.
  - They are abundance,
  - they are engineered knowledge for humans to consume,
  - they proved context, and provenance, and traceability
- Goal:
  - Our aim is to build institutional knowledge extracting pipeline that accumulates future intelligence and connect, archive and retrieve legacy tacit knowledge.
- How:
  - we want to engineer that presentation and meeting minutes (which concretize meta data and action speak [mandate]) for machines to consume.
- Ensuring:
  - Data quality while extracting intelligence
  - Data interoperability between projects, initiatives, and programs across the organization

Design process method

- Step 1: My own definition in words (no BFO/CCO lenses)
- Step 2: Build BPMN map

Following steps are Co-authored with Design Pattern Assistant(1) (AI)

- Step 3: Apply rules of Thumb
- Step 4: Lookup BFO/CCO lenses on to build Design pattern
- Step 5: Pattern Design workshop Mermaid Draw

(1) Design Pattern Assistant is an agent that is trained on BFO and CCO, it's job it identify and re-use entity classification and relationships. A sample of their assistant will be provided in markdown files

Step 1: My own definition, tested

- **v1:** A Meeting is a type of human gathering (online/in person) for a determined period of time to address a topic(s) and make decisions (mandates)

Test (Military Intelligence Pt. 1)	Counterexample	Result
Term true, definition false (sl. 5)	A status briefing where nothing is decided	**Fails:** "make decisions" isn't true of every meeting
Term true, definition false (sl. 5)	An unscheduled 5-minute corridor huddle	**Fails:** "a determined period of time" isn't either
Definition true, term false (sl. 5)	A family at dinner deciding on a holiday	**Too broad:** nothing requires a planned act or common purpose
Avoid negative language (sl. 12–27)	—	**Passes.** Keep it that way: define a subtype, not a "non-decision meeting"
Word-to-world (sl. 30–32)	"make decisions (mandates)"	**Conflates** an act (deciding) with its output (the mandate)

A definition is a pair <T, E>: T is true of a domain just in case E is (Military Intelligence Part 1, slide 4).

Step 1 revised: v2 definitions

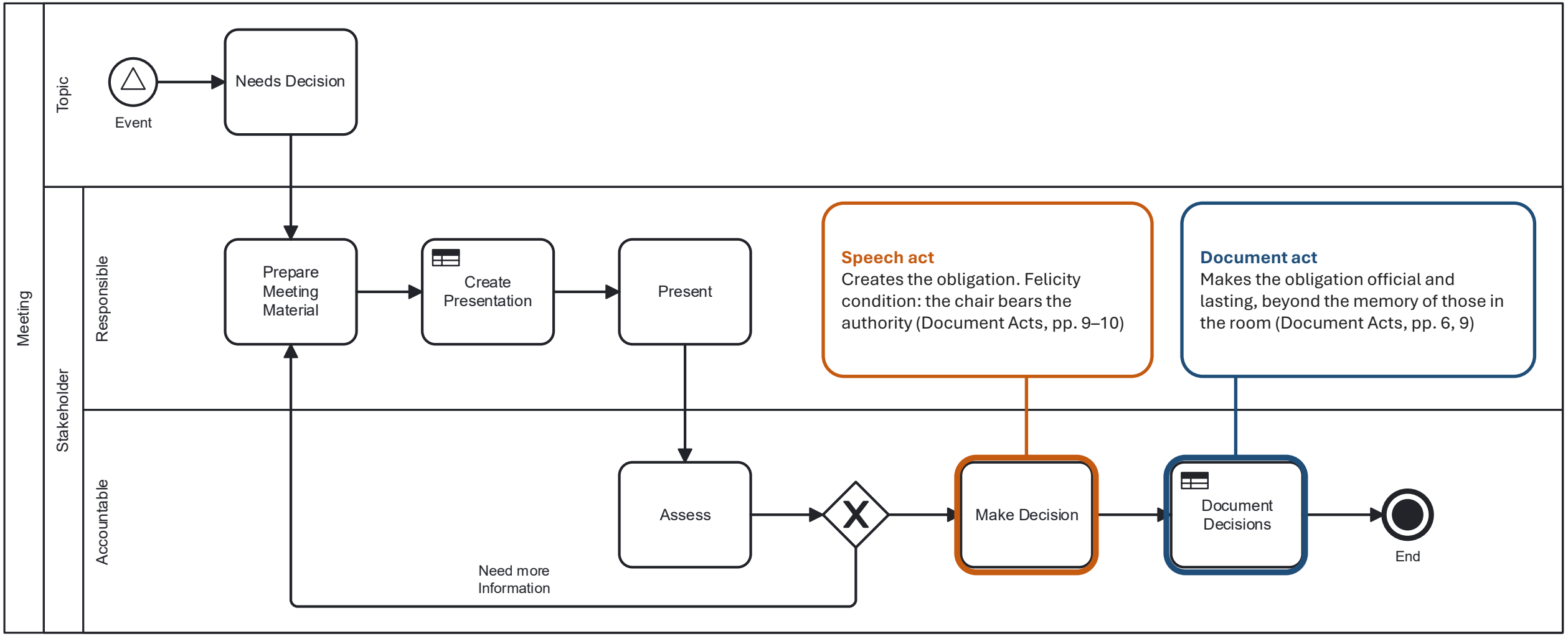
- **Meeting** =def an act of association in which two or more persons, sharing a place or connected through a communication artifact, communicate about one or more topics for a common purpose
- **Decision meeting** =def a meeting that has as part an act of directive communication, by a person bearing an authority role, whose output is a mandate
- **Mandate** =def a process regulation that is the output of an act of directive communication in a meeting

Case	Meeting?	Decision mtg?
Status briefing, nothing decided	✓	✗ (correct)
Corridor huddle that settles an issue	✓	✓
Family dinner deciding a holiday	✗	—
Board meeting, chair issues a directive	✓	✓

What changed: genus from CCO Act of Meeting ("...assemble for some common purpose"). "Determined period" dropped: every meeting takes time, but not every one is scheduled. Deciding moved to a subtype, so the class is split from the inside with no negative definition (sl. 21–24).

Open: "common purpose" carries a lot of weight.

Step 2: Build BPMN map (two acts, not one)



Step 3: Apply rules of Thumb

- Slot 1: Material Entity
- Slot 2: Quality
- Slot 3: Process
- Slot 4: Realizable Entity
- Slot 5: Where
- Slot 6: When
- Slot 7: Information

Exercise 3: Meeting · Step 1: Identify

Inputs: your definition (narrative) + `Meeting.bpmn` (takes precedence where they disagree) **Mode:** run straight through with no pauses, as you asked. Where I'd normally ask a question, I've recorded an **assumption (A1–A7)** instead. Override any of them and the pattern updates. **Scope:** no competency question, so the scope is the BPMN plus the narrative **Level:** both. A class-level spec for "a type of human gathering…", plus one typed example run of the BPMN **Parser output:** `skeleton.md` / `skeleton.tsv`, passes lint **Related:** your Final Presentation mandate pattern (`01-cdp-mandate-pattern.md`), which models the "make decisions (mandates)" half

Source key: **B:** BPMN element · **N:** narrative phrase · **implied**

Slot 1: Material Entity

Item	Source	Note
the stakeholders (the attendees)	B: lane *Stakeholder* · N: "human gathering"	a group of persons
the responsible person	B: nested lane *Responsible*	prepares and presents
the accountable person	B: nested lane *Accountable*	assesses, decides, documents
files: material, deck, minutes	implied by B: *Create Presentation*, *Document Decisions* (business-rule tasks)	what the Slot 7 content is carried in
~~Topic~~	B: lane *Topic*	not something physical; it's **information**, so it moves to Slot 7 (A3)

Step 4: Lookup BFO/CCO lenses on to build Design pattern

- Every term below was looked up in the index built from **CCO v2.2 (2026-08-13)**, which includes BFO 2020.
- **Marks:** ✓ exact CCO match · ~ near match ·
  - proposed new term · **— stays at BFO** means CCO has nothing and the pattern doesn't need anything more specific

Exercise 3: Meeting · Step 2: Map to BFO, then CCO

Every term below was looked up in the index built from CCO v2.2 (2026-08-13), which includes BFO 2020. Your Pause 1 answers are applied. Items marked M1–M7 are new mapping decisions for you to challenge at Pause 2.

Marks: ✓ exact CCO match · ~ near match · * proposed new term · **— stays at BFO** means CCO has nothing and the pattern doesn't need anything more specific

2.1 BFO

Slot	Item	BFO class	Why
1	the stakeholders	`object aggregate`	separate persons that together form a unit
1	responsible person · accountable person · files	`object`	single physical things; a file is carried by a physical medium
3	triggering event · every act · the meeting · the cycle	`process`	
4	the accountable person's authority	`role`	exists because others accept the person's power to issue binding directives, not because of the person's physical makeup
5	the meeting room	`site`	an area of space, not made of matter, bounded by walls

Steps 5 the workshop

- We (DPA and I) take the inputs from Steps 3, 4
- Conduct a workshop to build the design pattern

Exercise 3: Meeting · Step 3: Workshop

Built from: `Meeting.bpmn` (which takes precedence) and your definition · assumptions A1–A7 from Step 1 (no pauses) **Size:** 63 triples, 33 nodes · **Lint:** PASS, 0 errors · **New terms:** none **Terms and reasons:** `step2.md` · **Machine-readable:** `triples.tsv`

How to read: oval = class · rectangle = instance, with its type in italics · every type is a verified `bfo:` or `cco:` term

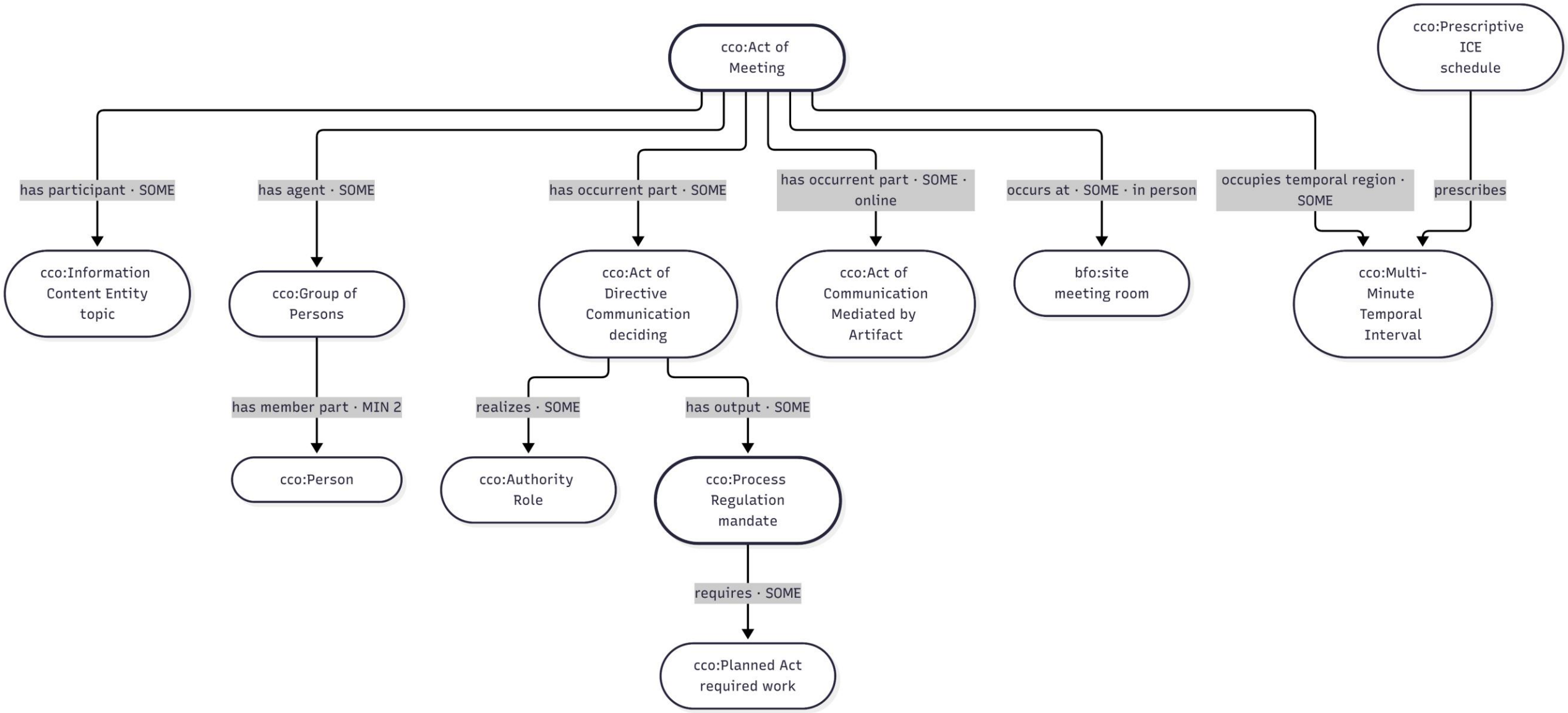
How to respond: override an assumption (`A2: …`) or reply to a decision (`D4: challenge — …`)

Panel A: your definition, clause by clause (class level)



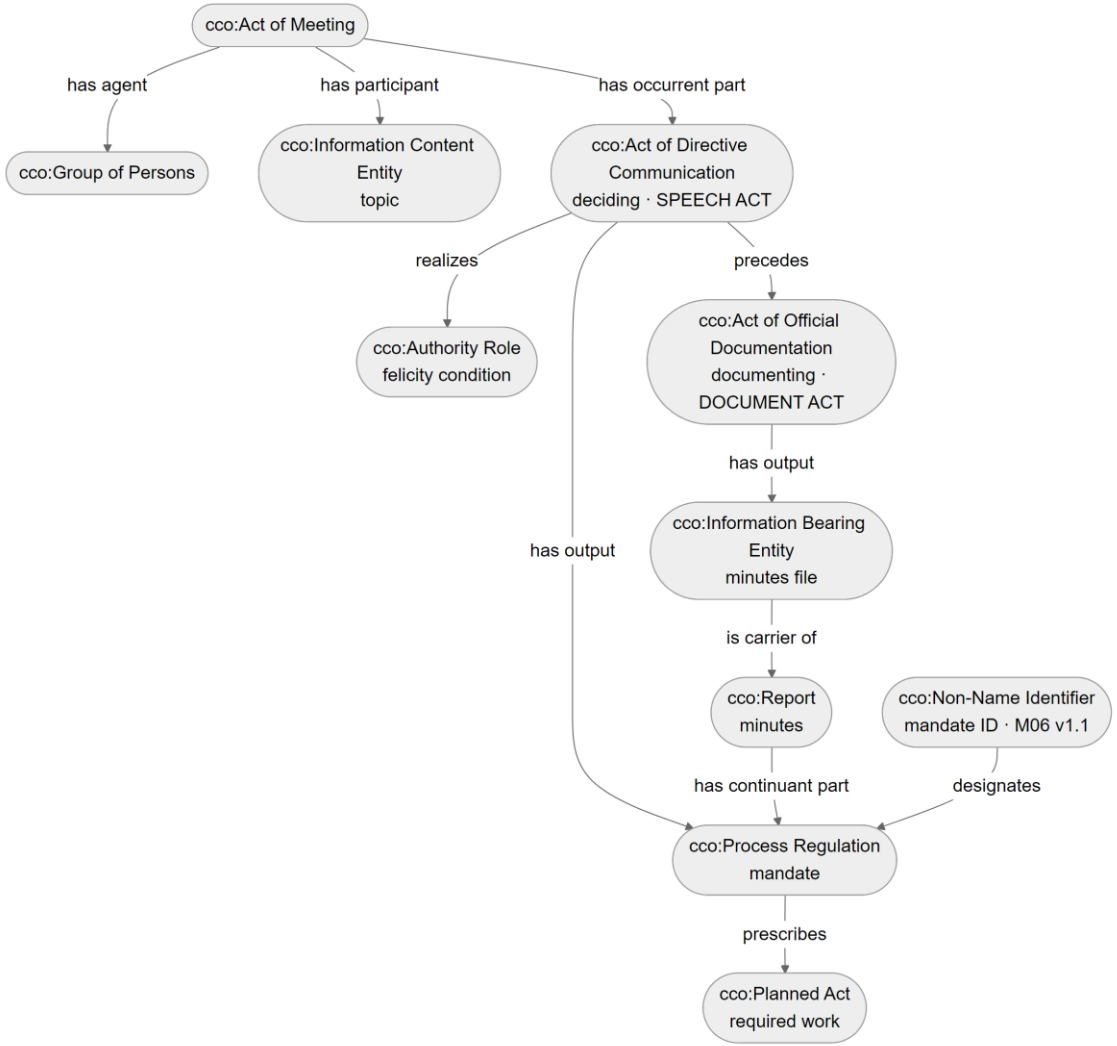
Your definition says	The pattern says
"a type of **human gathering**"	`Act of Meeting`: "two or more Persons assemble for some common purpose". Its agent is a `Group of Persons` with **at least 2** persons
"**(online / in person)**"	it happens *either* in some room (`occurs at` a site) *or* has parts where technology carries the communication (`Act of Communication Mediated by Artifact`)

Steps 5 Workshop result: Mermaid (v1)

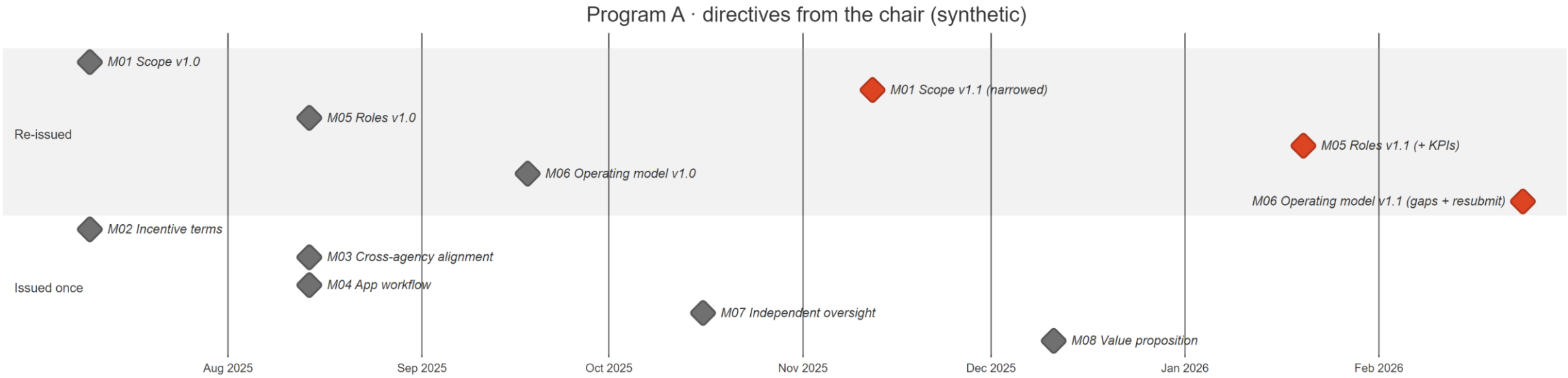


Step 5 revisited: speech act vs. document act

- **v1 problem:** one node, "deciding", did two jobs. The BPMN already had two tasks
- **Speech act:** cco:Act of Directive Communication realizes the Authority Role and outputs the mandate
- **Document act:** cco:Act of Official Documentation outputs the minutes file, which carries the minutes; the mandate is part of them
- **Identifier:** "M06 v1.1" designates the mandate, and the same ID links minutes, letters and trackers (Document Acts, p. 10)
- **Lint:** PASS, 0 errors, 3 warnings (justified)



Identifiers reveal recurrence



Document Acts (p. 10)	In this pattern
Tracking numbers and VINs create "virtual attachment between documents via cross-referencing"	A mandate ID (M06) appears in the minutes, a follow-up letter and a tracker: three documents, one mandate
The document's identifiers support knowledge by comparison	Comparing M06 v1.0 with v1.1 shows the chair raised it twice, with more asked the second time. Each version is a distinct mandate, linked by precedes

Synthetic data. The real program file is internal and is not published.

File Engineering aligned to IAO

CCO is imported; IAO is aligned (skos:closeMatch), not imported, so there is never a second "document" class. Following IAO-Intel, each file type is populated downward from IAO.

File element	Source label	IAO class (aligned)	CCO class (used)
Priority file (dossier)	ملف	document IAO_0000310	Information Content Entity
Meeting presentation	عرض	document IAO_0000310	Document Content Entity
Minutes	محضر	report IAO_0000088 (exact)	Report
Official letter	خطاب	document IAO_0000310	Document Content Entity
Directive / mandate	توجيه	directive information entity IAO_0000033	Process Regulation
Recording the decision	توثيق	documenting IAO_0000572	Act of Official Documentation
Mandate ID + version	M06 v1.1	identifier IAO_0020000	Non-Name Identifier
Executive priority	أولوية	objective specification IAO_0000005	Objective

People stay in CCO (IAO covers information only). IAO's version number is defined for manufactured products, so versions are part of the identifier instead. Precedent: IAO-Intel sets Intelligence Report under IAO Report (Smith et al., Table 1).

Course material applied in this revision

- **Beverley, Intelligence Analysis: Military Intelligence Part 1**: definitions as <T, E>, two-way counterexample tests, negative language, word-to-world fit → Step 1
- **Smith, Document Acts** (week 3): "to record" (p. 6); felicity conditions and documents as enduring deontic power (pp. 9–10); identifiers and knowledge by comparison (p. 10) → Steps 2 and 5
- **Smith et al., IAO-Intel** (week 3): downward population from document typologies; incremental coverage → File Engineering alignment
- **CCO v2.2 and IAO**: every term checked against the source files, with IRIs in the repository write-up
- **Done since:** the pattern in OWL (v0.1, 17 Oct), HermiT and CQ1–CQ5 pass. Next: sustained extendability, so every new document adds to it

Next step: sustained extendability

Every new presentation or set of minutes should make the base deeper or wider, and show how the organization itself names what it knows.

Depth

New facts about things the base already holds

D05 re-issues M01, M05, M06. The new acts are preceded by acts from D01 and D02, which are never edited.

Breadth

A new branch, only when a question needs it

D03, the first deck, brings in the objectives and links them to work the minutes already hold.

Language

The organization's own words for what is already there

محضر. MoM → minutes

تكليف. directive → mandate

“KPI” parked as a SKOS concept

Sustained extendability =def *a property of a knowledge base built from an organization's documents, such that each new document passes through one fixed loop that* ***links*** *what it contributes to what the base already holds, as depth, breadth or language,* ***while every earlier assertion and every earlier answer is kept***.

Synthetic data (Program A). Documents D01–D07 are in final-project/ontology/documents/.

Defined and tested like “meeting”

v1: *each new document can be added to the base without changing earlier assertions*

Test (Military Intelligence Pt. 1)	Counterexample	Result
Definition true, term false	A document store or search index: every file added, nothing earlier changed, nothing integrated	**Fails:** adding is not extending
Term true, definition false	Later minutes say M06 was re-issued: something new is known, so the base must change	**Fails:** the problem is overwriting, not changing
Avoid negative language	“without changing”	**Restate** as what is kept

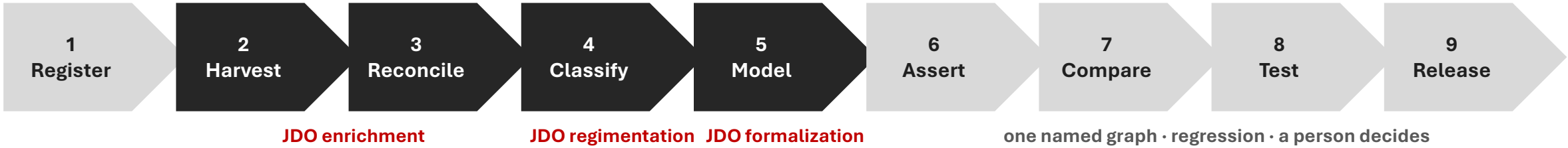
v2: each new document passes through **one fixed loop** that **links** what it contributes to what the base already holds, while **every earlier assertion and every earlier answer is kept**

“Links” rules out the document store: measured as links into the base

“Every earlier answer” is a test I can run: CQ1–CQ5 re-run after every document

“One fixed loop” is what makes it sustained: document 10 goes through the same steps as document 1

One loop per document, after JDO



Five rules keep it extendable

1. **One named graph per document**
   nothing merged in a way that loses its source (Sikos)

2. **A new class only when a question needs it**
   otherwise a SKOS concept: “KPI” in D03

3. **Never overwrite: add, with a date**
   M06 v1.0 stays when v1.1 arrives (Document Acts, p. 10)

4. **Every new class under a BFO/CCO parent**
   downward population, genus + differentia (IAO-Intel)

5. **No IRI carries a project or file code**
   a class minted for one file must fit the next

Implemented: one document at a time

Document (synthetic)	Kind	Classes reused	New labels	CQs answered	Regressions
D01 minutes, meeting 1	base	0/11	0	4/5	—
D02 minutes, meeting 2	depth	8/8	1	4/5	0
D03 deck, meeting 3	breadth	1/3	6	4/5	0
D04 minutes, meeting 3	depth	4/5	2	4/5	0
D05 minutes, meeting 4	depth	8/8	0	**5/5 (CQ1)**	0
D06 follow-up letter	language	1/2	5	5/5	0
D07 conflicting note	**rejected**	—	—	—	—

0 new OWL classes in six documents. “KPI” stayed a SKOS concept

0 regressions. After D06, CQ1–CQ5 equal the one-shot v0.1 answers

1 document rejected by HermiT. The base stays at D06

python ingest.py: merge base + accepted documents + new one → HermiT → CQ1–CQ5 → every earlier answer must still hold → count what was added. Log: final-project/ontology/results/extendability/ingest-log.md

Named graphs, and a rejected document

Question (named graphs)	Program A answer
X01: which documents is each mandate known from?	M06 v1.1: minutes D05 and letter D06. All others: one set of minutes
X04: which mandates rest on one source?	6 of 7. Only M06 v1.1 is corroborated
X02: what did D05 add to what was known?	6 facts on old nodes: 3 × precedes, 3 × prescribes. Nothing overwritten
V04: which words did each document add?	14 labels, Arabic and English, across D02, D03, D04, D06

D07 says: *"M06 v1.1 is a Planned Act"*

minutes_4 has continuant part M06 v1.1

→ M06 v1.1 is a continuant (range)

Planned Act ⊑ Act ⊑ process ⊑ occurrent

continuant disjointWith occurrent (BFO)

→ inconsistent: rejected

The mandate was confused with the work it requires. The base stays at D06.

CQ1 first answers at D05. Recurrence is invisible inside any one document. It appears only when a later document is linked to earlier ones, which is why the documents have to extend one base.

Course material applied to extendability

Joint Doctrine Ontology (week 7, §IX.C): built incrementally, one publication at a time; enrichment → regimentation → formalization → the loop

Ontology for the Intelligence Analyst (week 1): terms keep their place "even when … whole new branches are added"; one preferred label per type → language goes to labels

IAO-Intel (week 3, §V): coverage "determined incrementally" from requests and document terms → a class only when a question needs it

Document Acts (week 3): identifiers create virtual attachment → D06 links to M06 v1.1 without pointing at D05

C3O (week 11) and Sikos, Cybersecurity Knowledge Graphs (week 11): domain modules under CCO; one named graph per source

Falsehoods Programmers Believe About Names (week 13): names are not unique or stable → محضر, محضر اجتماع, MoM: one concept

Limits: synthetic and small, so it shows the mechanism, not a trend · provenance stops at the document (real minutes need page + quote) · the regression test protects only what the CQs ask

Next: the final project packet; run the same loop on a real file, where the rate of new terms per document should fall as the vocabulary matures